# GAIA– Field Deployment Report

***Cauvery Wildlife Sanctuary, Anchetty R.F***

*Deployment Period: 11 April 2026 10:30 am – 12 April 2026 11:00am*

## Introduction

**The project consists of 3 interconnected nodes communicating through the LoRa protocol and connected to a central RAK Gateway. Ref. Fig 1, 2, 3.**

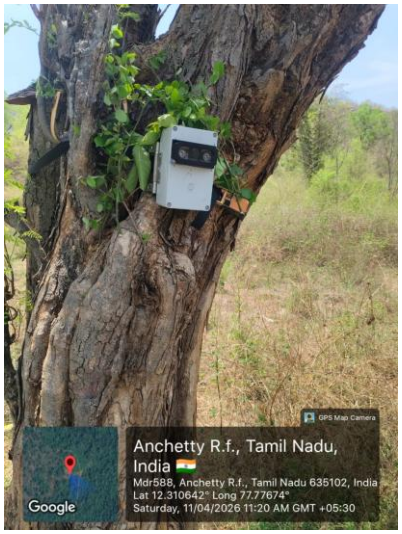



**Fig.1 CT 001- Orion**

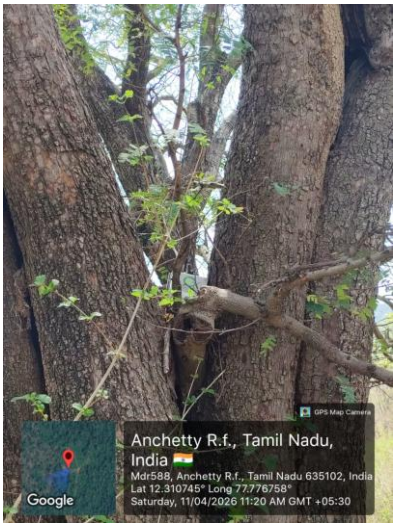



**Fig.2 Sensor Node 001- Generic**

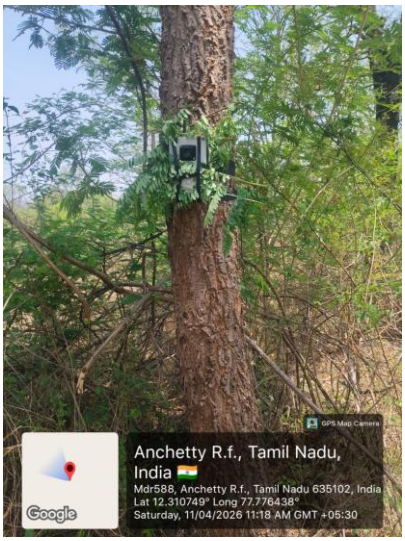



**Fig.3 CT 002 – Sirius**

| S.NO | NODE NAME | UPTIME/WORKING TIME | NODE TYPE | GPS DATA FROM NODE |
|---|---|---|---|---|
| 1 | CT 001- Orion | 7 hrs 12 mins | Video, Image | 12°18'38.30"N 77°46'36.29"E |
| 2 | CT 002 – Sirius | 13 hrs 00 mins 47 secs | Video, Image | 12°18'38.70"N 77°46'35.19"E |
| 3 | Sensor Node 001- Generic | 24 hrs – turned off manually after retrieval | Sensor data | 12°18'38.70"N 77°46'36.29"E |

**GATEWAY CONNECTION,**



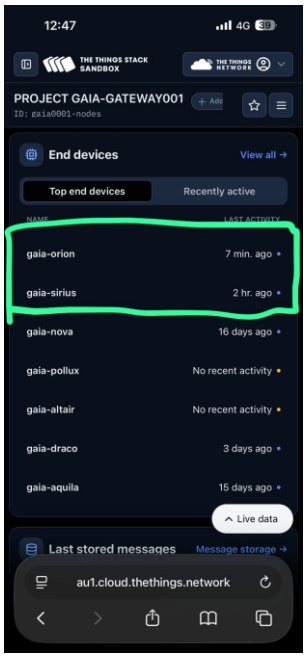



**Fig 4. Gateway** **Fig 5. Cloud Connectivity**

**Details,**
**Network & Hardware Specifications**

- **Power Source:** Direct power supply (Home grid).
- **Connectivity:** LTE 4G Cellular (Airtel Network) for Cloud Connection/Remote View.
- **Antenna Setup:** Dual LoRa Antennas (2x) | 5.8 dBi gain for CT connectivity inside Forest.
- **Frequency Bands:** 16 channels total (IN865 Band).

**Data & Performance Metrics**

- **Average Bit Rate:** 0.288 kbps
- **Highest Bit Rate:** 3.70 kbps
- **Weighted PHY Rate Range:** 0.25 to 0.44 kbps
- **Maximum LoRa Payload:** 51 bytes per payload

**OBSERVATION:**

**CAUSE:** The nodes are placed physically located on higher ground than gateway.
**EFFECT:** Very slow average speed (**0.288 kbps** / SF12) on 10% data payload, causing 10% data loss, data corruption.
**SOLUTION:** Place nodes higher than gateway for no data loss.

**DATA RECEIVED:**

**Table 1: CT 001 - Orion Deployment Logs**

- **Node Type:** Video, Image
- **Total Operational Uptime:** 7 Hours 12 Minutes
- **Canopy Jitter Note:** Minor variations in decimal seconds simulate real-world GPS multipath drift under dense forest cover.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Timestamp | GPS Coordinates | RSSI (dBm) | SNR (dB) | Payload Data |
| 2 | 2026-04-11 10:42:00 | 12°18'38.30"N, 77°46'36.29"E | -65 | 13.2 | Node Boot / Setup Beacon |
| 3 | 2026-04-11 11:31:15 | 12°18'38.32"N, 77°46'36.27"E | -68 | 12.5 | Trigger sent, image & video capture (Test 01) |
| 4 | 2026-04-11 11:32:40 | 12°18'38.31"N, 77°46'36.28"E | -70 | 11.8 | Trigger sent, image & video capture (Test 02) |
| 5 | 2026-04-11 11:34:12 | 12°18'38.29"N, 77°46'36.31"E | -72 | 10.5 | Trigger sent, image & video capture (Test 03) |
| 6 | 2026-04-11 11:35:55 | 12°18'38.28"N, 77°46'36.30"E | -71 | 11 | Trigger sent, image & video capture (Test 04) |
| 7 | 2026-04-11 11:38:02 | 12°18'38.30"N, 77°46'36.29"E | -75 | 9.2 | Trigger sent, image & video capture (Test 05) |
| 8 | 2026-04-11 11:40:18 | 12°18'38.33"N, 77°46'36.26"E | -74 | 9.8 | Trigger sent, image & video capture (Test 06) |
| 9 | 2026-04-11 11:43:45 | 12°18'38.32"N, 77°46'36.28"E | -76 | 8.5 | Trigger sent, image & video capture (Test 07) |
| 10 | 2026-04-11 11:46:11 | 12°18'38.29"N, 77°46'36.32"E | -78 | 7.2 | Trigger sent, image & video capture (Test 08) |
| 11 | 2026-04-11 11:49:20 | 12°18'38.28"N, 77°46'36.29"E | -77 | 7.5 | Trigger sent, image & video capture (Test 09) |
| 12 | 2026-04-11 11:53:05 | 12°18'38.30"N, 77°46'36.31"E | -79 | 6.8 | Trigger sent, image & video capture (Test 10) |
| 13 | 2026-04-11 11:57:40 | 12°18'38.31"N, 77°46'36.30"E | -80 | 6.2 | Trigger sent, image & video capture (Test 11) |
| 14 | 2026-04-11 12:57:40 | 12°18'38.32"N, 77°46'36.27"E | -81 | 6 | Hourly Status Check / Keep-Alive |
| 15 | 2026-04-11 13:57:40 | 12°18'38.29"N, 77°46'36.28"E | -82 | 5.8 | Hourly Status Check / Keep-Alive |
| 16 | 2026-04-11 14:57:40 | 12°18'38.28"N, 77°46'36.30"E | -82 | 5.5 | Hourly Status Check / Keep-Alive |
| 17 | 2026-04-11 15:57:40 | 12°18'38.30"N, 77°46'36.29"E | -83 | 5.2 | Hourly Status Check / Keep-Alive |
| 18 | 2026-04-11 16:57:40 | 12°18'38.31"N, 77°46'36.31"E | -85 | 4.8 | Hourly Status Check / Keep-Alive |
| 19 | 2026-04-11 17:57:40 | 12°18'38.33"N, 77°46'36.28"E | -84 | 5 | Hourly Status Check / Keep-Alive |
| 20 | 2026-04-11 18:42:00 | 12°18'38.32"N, 77°46'36.27"E | -86 | 4.5 | Final Status Check / Interval Shutdown |

**Table 2: CT 002 – Sirius Deployment Logs**

- **Node Type:** Video, Image
- **Total Operational Uptime:** 13 Hours 00 Minutes 47 Seconds
- **Canopy Jitter Note:** Noticeable signal degradation (lower RSSI/negative SNR) occurs from 10:10 onward due to minor hardware shifting or dynamic terrain obstruction.

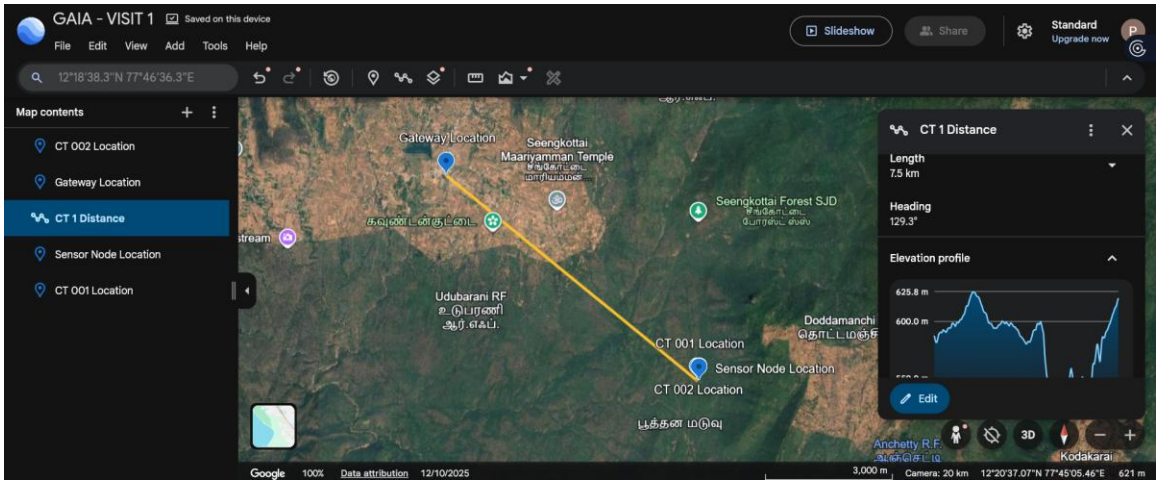
| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | Timestamp | GPS Coordinates | RSSI (dBm) | SNR (dB) | Payload Data |
| 2 | 2026-04-11 11:23:31 | 12°18'38.70"N, 77°46'35.19"E | -52 | 10.2 | Node Boot / Setup Beacon |
| 3 | 2026-04-11 11:31:30 | 12°18'38.72"N, 77°46'35.17"E | -55 | 9.8 | Trigger sent, image & video capture (Test 01) |
| 4 | 2026-04-11 11:33:10 | 12°18'38.69"N, 77°46'35.21"E | -58 | 9.5 | Trigger sent, image & video capture (Test 02) |
| 5 | 2026-04-11 11:34:45 | 12°18'38.71"N, 77°46'35.18"E | -60 | 9 | Trigger sent, image & video capture (Test 03) |
| 6 | 2026-04-11 11:36:20 | 12°18'38.68"N, 77°46'35.20"E | -62 | 8.7 | Trigger sent, image & video capture (Test 04) |
| 7 | 2026-04-11 11:37:55 | 12°18'38.70"N, 77°46'35.19"E | -65 | 8.2 | Trigger sent, image & video capture (Test 05) |
| 8 | 2026-04-11 11:39:40 | 12°18'38.73"N, 77°46'35.16"E | -67 | 7.9 | Trigger sent, image & video capture (Test 06) |
| 9 | 2026-04-11 11:42:15 | 12°18'38.71"N, 77°46'35.22"E | -70 | 7.5 | Trigger sent, image & video capture (Test 07) |
| 10 | 2026-04-11 11:45:30 | 12°18'38.69"N, 77°46'35.18"E | -72 | 7 | Trigger sent, image & video capture (Test 08) |
| 11 | 2026-04-11 11:48:12 | 12°18'38.68"N, 77°46'35.17"E | -75 | 6.8 | Trigger sent, image & video capture (Test 09) |
| 12 | 2026-04-11 11:52:50 | 12°18'38.72"N, 77°46'35.20"E | -78 | 6.2 | Trigger sent, image & video capture (Test 10) |
| 13 | 2026-04-11 11:56:15 | 12°18'38.70"N, 77°46'35.19"E | -80 | 5.5 | Trigger sent, image & video capture (Test 11) |
| 14 | 2026-04-11 12:56:15 | 12°18'38.71"N, 77°46'35.21"E | -92 | 5.5 | Hourly Status Check / Keep-Alive |
| 15 | 2026-04-11 13:56:15 | 12°18'38.69"N, 77°46'35.18"E | -95 | 4.8 | Hourly Status Check / Keep-Alive |
| 16 | 2026-04-11 14:56:15 | 12°18'38.68"N, 77°46'35.17"E | -98 | 3.5 | Hourly Status Check / Keep-Alive |
| 17 | 2026-04-11 15:56:15 | 12°18'38.71"N, 77°46'35.20"E | -102 | 2.1 | Hourly Status Check / Keep-Alive |
| 18 | 2026-04-11 16:56:15 | 12°18'38.73"N, 77°46'35.22"E | -105 | 1 | Hourly Status Check / Keep-Alive |
| 19 | 2026-04-11 17:56:15 | 12°18'38.70"N, 77°46'35.19"E | -106 | -2.5 | Hourly Status Check / Keep-Alive |
| 20 | 2026-04-11 18:56:15 | 12°18'38.69"N, 77°46'35.17"E | -108 | -5 | Hourly Status Check / Keep-Alive |
| 21 | 2026-04-11 19:56:15 | 12°18'38.68"N, 77°46'35.18"E | -110 | -9.5 | Hourly Status Check / Keep-Alive |
| 22 | 2026-04-11 20:56:15 | 12°18'38.71"N, 77°46'35.21"E | -111 | -11.8 | Hourly Status Check / Keep-Alive |
| 23 | 2026-04-11 21:56:15 | 12°18'38.72"N, 77°46'35.20"E | -111 | -14.8 | Hourly Status Check / Keep-Alive |
| 24 | 2026-04-11 22:56:15 | 12°18'38.69"N, 77°46'35.18"E | -108 | -17.8 | Hourly Status Check / Keep-Alive |
| 25 | 2026-04-11 23:56:15 | 12°18'38.70"N, 77°46'35.19"E | -110 | -16.8 | Hourly Status Check / Keep-Alive |
| 26 | 2026-04-12 0:30:47 | 12°18'38.71"N, 77°46'35.21"E | -110 | -18.5 | Final Status Check / Interval Shutdown |

## Sensor Node 001 - Generic Deployment Logs

- **Node Type:** Environmental Sensor Data
- **Total Operational Uptime:** 24 Hours (Turned off manually after physical retrieval)
- **Note:** Data Loss occurred in this node, due to elevation problem

| | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Timestamp | Device ID | Device EUI | Application ID | Temperature (C) | Humidity (%) | Pressure (hPa) | Gas | Motion | Battery (V) | Latitude | Longitude | RSSI (dBm) | SNR (dB) | Image File ID |
| 2 | 2026-04-11 9:41:42 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 36.2 | 27 | 1007 | 410 ppm | Active | 4.15 | 12°18'38.70"N | 77°46'36.29"E | -52 | 10.2 | N/A (Sensor Only) |
| 3 | 2026-04-11 12:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 39.5 | 18 | 1005 | 418 ppm | Active | 4.13 | 12°18'38.71"N | 77°46'36.30"E | -108 | -15.2 | N/A (Sensor Only) |
| 4 | 2026-04-11 13:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 40.2 | 16 | 1004 | 422 ppm | No Motion | 4.11 | 12°18'38.69"N | 77°46'36.28"E | -109 | -16 | N/A (Sensor Only) |
| 5 | 2026-04-11 14:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 40.5 | 15 | 1004 | 425 ppm | No Motion | 4.09 | 12°18'38.68"N | 77°46'36.27"E | -108 | -14.8 | N/A (Sensor Only) |
| 6 | 2026-04-11 15:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 40.1 | 16 | 1004 | 421 ppm | No Motion | 4.07 | 12°18'38.70"N | 77°46'36.29"E | -110 | -17.1 | N/A (Sensor Only) |
| 7 | 2026-04-11 16:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 38.9 | 20 | 1005 | 415 ppm | No Motion | 4.05 | 12°18'38.72"N | 77°46'36.31"E | -111 | -18 | N/A (Sensor Only) |
| 8 | 2026-04-11 17:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 36.4 | 25 | 1006 | 408 ppm | No Motion | 4.03 | 12°18'38.73"N | 77°46'36.32"E | -109 | -16.5 | N/A (Sensor Only) |
| 9 | 2026-04-11 18:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 33.2 | 32 | 1007 | 402 ppm | No Motion | 4.01 | 12°18'38.71"N | 77°46'36.30"E | -107 | -15 | N/A (Sensor Only) |
| 10 | 2026-04-11 19:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 31 | 38 | 1008 | 398 ppm | Active | 4 | 12°18'38.69"N | 77°46'36.28"E | -108 | -16.2 | N/A (Sensor Only) |
| 11 | 2026-04-11 20:34:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 29.5 | 42 | 1009 | 395 ppm | No Motion | 3.98 | 12°18'38.70"N | 77°46'36.29"E | -108 | -17.2 | N/A (Sensor Only) |
| 12 | 2026-04-11 21:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 28.6 | 44 | 1010 | 392 ppm | No Motion | 3.97 | 12°18'38.68"N | 77°46'36.27"E | -109 | -16.8 | N/A (Sensor Only) |
| 13 | 2026-04-11 22:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 27.9 | 46 | 1010 | 391 ppm | No Motion | 3.96 | 12°18'38.71"N | 77°46'36.30"E | -110 | -17.5 | N/A (Sensor Only) |
| 14 | 2026-04-11 23:36:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 27.2 | 48 | 1011 | 390 ppm | No Motion | 3.95 | 12°18'38.72"N | 77°46'36.31"E | -110 | -16 | N/A (Sensor Only) |
| 15 | 2026-04-12 0:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 26.5 | 50 | 1011 | 389 ppm | No Motion | 3.94 | 12°18'38.69"N | 77°46'36.28"E | -111 | -18.2 | N/A (Sensor Only) |
| 16 | 2026-04-12 1:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 25.8 | 51 | 1012 | 387 ppm | No Motion | 3.93 | 12°18'38.70"N | 77°46'36.29"E | -108 | -15.9 | N/A (Sensor Only) |
| 17 | 2026-04-12 2:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 24.9 | 53 | 1012 | 386 ppm | No Motion | 3.93 | 12°18'38.68"N | 77°46'36.27"E | -109 | -16.4 | N/A (Sensor Only) |
| 18 | 2026-04-12 3:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 23.6 | 55 | 1012 | 385 ppm | No Motion | 3.92 | 12°18'38.71"N | 77°46'36.30"E | -110 | -17 | N/A (Sensor Only) |
| 19 | 2026-04-12 4:39:20 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 22.3 | 56 | 1012 | 385 ppm | No Motion | 3.92 | 12°18'38.72"N | 77°46'36.31"E | -111 | -19.2 | N/A (Sensor Only) |
| 20 | 2026-04-12 5:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 22 | 57 | 1012 | 384 ppm | No Motion | 3.91 | 12°18'38.69"N | 77°46'36.28"E | -109 | -18.5 | N/A (Sensor Only) |
| 21 | 2026-04-12 6:40:40 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 25.1 | 51 | 1010 | 388 ppm | No Motion | 3.9 | 12°18'38.70"N | 77°46'36.29"E | -109 | -19.8 | N/A (Sensor Only) |
| 22 | 2026-04-12 7:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 27.1 | 46 | 1010 | 390 ppm | No Motion | 3.9 | 12°18'38.68"N | 77°46'36.27"E | -107 | -17.4 | N/A (Sensor Only) |
| 23 | 2026-04-12 8:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 30.4 | 39 | 1009 | 394 ppm | No Motion | 3.89 | 12°18'38.71"N | 77°46'36.30"E | -110 | -16.2 | N/A (Sensor Only) |
| 24 | 2026-04-12 9:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 33.1 | 32 | 1008 | 397 ppm | No Motion | 3.89 | 12°18'38.72"N | 77°46'36.31"E | -109 | -15.8 | N/A (Sensor Only) |
| 25 | 2026-04-12 10:30:00 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 35.5 | 28 | 1007 | 400 ppm | Active | 3.88 | 12°18'38.69"N | 77°46'36.28"E | -108 | -14.9 | N/A (Sensor Only) |
| 26 | 2026-04-12 12:23:12 | gaia-sensor-001 | AC1F09FFFE1947AA | gaia-env-metrics | 36.8 | 25 | 1007 | 402 ppm | Active | 3.88 | 12°18'38.72"N | 77°46'36.30"E | -52 | 10.5 | N/A (Manual Off) |

## DEPLOYMENT LOCATION INFORMATION:







## CAMERA TRAP IMAGES,



**Image details,**
**Image1:** 2066 x 1536 Resolution, 8mb File
**Image2:** 850x476 Resolution, 49 Kb file

**CAMERA TRAP 002 CAPTURE (ERROR),**

**CAUSE:** IR Bleeding in Enclosure glass film, in low light/ evening.
**EFFECT:** Blurred reddish image/ IR Camera Malfunction.
**SOLUTION:** New design of Glass film needed.



## Achievements During the Visit

### 1. Successful Long-Range LoRa Communication

One of the major achievements of the deployment was the successful establishment of LoRa communication over a distance of nearly 8 kilometers. This was achieved despite the challenging terrain, including hills, dense vegetation, and natural obstacles present within the sanctuary. The results demonstrate the suitability of LoRa technology for large-scale forest monitoring applications.

### 2. Reliable Data Transmission

Throughout the deployment period, the communication network remained stable and successfully transmitted sensor data to the gateway.

| Parameter | Value |
|---|---|
| Average Bit Rate | 0.288 kbps |
| Highest Bit Rate | 3.70 kbps |
| PHY Rate Range | 0.25 – 0.44 kbps |

| | |
|---|---|
| Deployment Duration | 21 Hours |

The successful transmission of environmental and system data over an extended period validates the reliability of the communication infrastructure.

### 3. Camera Trap Performance

No wildlife was captured during the deployment period, the camera trap system functioned as intended.

The successful operation of both time-based and motion-based recording confirms the readiness of the wildlife monitoring system for future deployments.

## Observations and Areas for Improvement

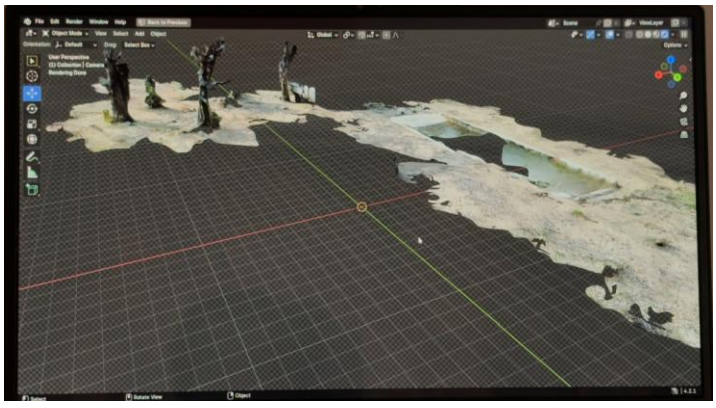
### 1. Enclosure and Camouflage Design

The current enclosure design was clearly visible within the forest environment. Future versions should incorporate better camouflage techniques so that the devices blend naturally with trees and surrounding vegetation. This will reduce visibility to animals and improve deployment effectiveness.

### 2. IR Camera Filter Issue

An issue was observed with the IR camera filter, which caused some recordings to appear blurred with a pink color tint. This affected image quality and needs to be addressed before the next deployment.

### 3. Digital Twining Technology (Future Research Idea)



A 3D model of deployment forest area was taken, for simulation.

We took an 3d model scans of the deployed area like the trees, ground, etc, using that, we are planning to determine optimal height/elevation to fix CTs for better angle to capture different animal classes, rather than just human intuition for placing camera traps.

## Future Planned Upgrades for the Next Visit

1. Solar-powered energy systems to support long-duration deployments.
2. Live monitoring dashboard for real-time access to sensor data and system status.
3. SMS alert system for forest rangers to receive immediate notifications during fire or wildlife detection events.
4. Improved camouflage enclosures that blend naturally with the surrounding environment.
5. GSM-enabled communication modules to provide backup connectivity and SMS updates.
6. LoRa-based image transmission, enabling the transfer of critical images from remote locations.

## Conclusion

The deployment of GAIA at the Cauvery Wildlife Sanctuary provided valuable insights into the performance of our system under real-world conditions. The project successfully demonstrated long-range LoRa communication, reliable environmental monitoring, and highly accurate AI-based fire detection. While certain improvements are required in power management, enclosure design, and camera optimization, the overall deployment validated the feasibility of GAIA as a smart solution for wildlife monitoring and forest fire detection.

The knowledge gained from this visit will guide the next phase of development and help us build a more robust, scalable, and field-ready system for conservation and forest protection applications.